

NAVIGATE THE FUTURE OF CODING

WILL AI REPLACE JUNIOR PROGRAMMERS?

PARKER HENDRIX

In just seconds, AI can now produce code that used to take developers hours to write—redefining what it means to be efficient in software development.

For junior developers like me entering the field, this is both exciting and unsettling. If AI can already write code in seconds, it raises concerns about job security. It's hard not to wonder whether the skills we're learning today will still hold value tomorrow or if AI will move so fast that we're left behind before we even get started. With AI now handling many foundational tasks, we have fewer opportunities to practice and build skills through meaningful coding experience.

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¹ Junior developers, in particular, are in trouble. Tasks that once provided valuable early-career experience are now handled by AI.
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Figure 1 illustrates the total number of jobs impacted by AI, along with the number of men and women affected. The data shows just how significantly AI is reshaping the workforce, with tens of millions of roles across the U.S. now highly exposed to generative AI automation.² AI isn't a distant threat—it is already transforming the way we work. With automation accelerating, it's more important than ever to understand AI's capabilities, stay adaptable, and navigate the future with confidence.

FIGURE 1: NUMBER OF JOBS IMPACTED BY AI

6 out of 10 Men



48,620,000

8 out of 10 Women



58,870,000

Source: Kenin Institute of Private Enterprise



THE EVOLUTION OF CODING WITH AI

Artificial intelligence is changing the software development world by streamlining workflows, improving code quality, and making collaboration easier across teams. Developers are now able to write code faster and with fewer errors, transforming how software is built and maintained. According to insights from the McKinsey Global Institute, AI and automation will continue to reshape the labor market, signaling that these tools are not just enhancements—but essential to staying competitive in the industry.³

Faster Development

AI coding tools significantly accelerate the software development process by creating code snippets, completing functions, and even producing entire modules from simple instructions. This frees developers from repetitive tasks, allowing them to focus more on solving complex problems and driving innovation. As a result, projects move from initial ideas to deployment quicker than before.

Fewer Bugs

For developers, dealing with bugs is a constant cycle of trial and error—one that AI tools help simplify and accelerate. AI analyzes code in real time to identify potential errors before they cause problems. It can suggest fixes, optimize logic, and even automatically generate test cases to catch hidden issues. This proactive approach results in more stable, reliable software and reduces the time spent debugging after deployment.

Better Learning

AI is one of the fastest ways to learn and grasp difficult topics in software development. With access to vast resources, it can explain concepts and provide examples in the way that works best for each developer. This personalized guidance helps programmers build proficiency quickly and confidently take on more complex challenges.

Higher Productivity

AI not only helps developers write code faster but also boosts overall productivity by supporting various stages of the development process. It can assist with planning projects, brainstorming ideas, generating documentation, and adding clear comments to code. This allows developers to focus more on coding and less on those other tasks, helping teams deliver higher-quality software more efficiently.



Improved Collaboration

AI helps bridge communication gaps within development teams by providing concise summaries of complex code and highlighting key changes in updates. It can also suggest standardized language for commit messages and pull requests, making it easier for everyone to stay aligned. By promoting transparency and clarity, AI supports smoother teamwork and accelerates the integration of new members into projects.

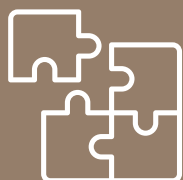
Though still relatively new in software development, AI is already making a significant impact. As it continues to evolve rapidly, its potential to reshape the way software is created and delivered is enormous, opening the door to even greater innovations and opportunities in the future. Junior developers need to understand AI's capabilities and be prepared to work alongside it to succeed in this evolving field.

- FASTER DEVELOPMENT
- FEWER BUGS
- BETTER LEARNING
- HIGHER PRODUCTIVITY
- IMPROVED COLLABORATION

READY TO ADAPT

As AI continues to evolve and integrate into nearly every aspect of work, the key to long-term success is learning how to grow alongside it. Andrew Ng, co-founder of Google Brain and Coursera, described artificial intelligence as “the new electricity”—a force powerful enough to transform every industry and generate new opportunities rather than eliminate them.⁴ Embracing this shift means recognizing that the most valuable skills are no longer just technical. It's about strengthening the uniquely human qualities that machines can't replace.





PROBLEM SOLVING



COMMUNICATION



AI LITERACY

Problem solving is still the foundation of effective development. Even when AI writes the code, humans must define the problem, outline the goal, and evaluate the results. Real-world challenges often have unclear requirements or conflicting needs. AI might suggest a solution, but only a person can judge whether it's the right one. Strong problem solving means thinking critically, making decisions, and adapting when things don't go as planned.

Communication remains a vital skill as technology evolves. Developers need to clearly express their ideas, justify technical decisions, and stay aligned with their teams and clients. While AI can support technical tasks, it can't replicate the clarity, empathy, or leadership that effective communicators contribute.

AI literacy is quickly becoming an essential skill for developers. You don't need to be an expert in how AI is built, but you should understand how it works, how to use it effectively, and how to critically assess its outputs. At its core, AI literacy involves the ability to engage with AI tools responsibly, apply them across different tasks, and recognize their broader impact.⁵ As these tools continue to grow in capability, those who know how to prompt, verify, and collaborate with AI will be the most valuable.

In this time of rapid change, the goal isn't to compete with AI but to complement it. By building skills that AI cannot replicate, you make yourself indispensable. Those who adapt will not only keep pace with innovation but help shape the future.



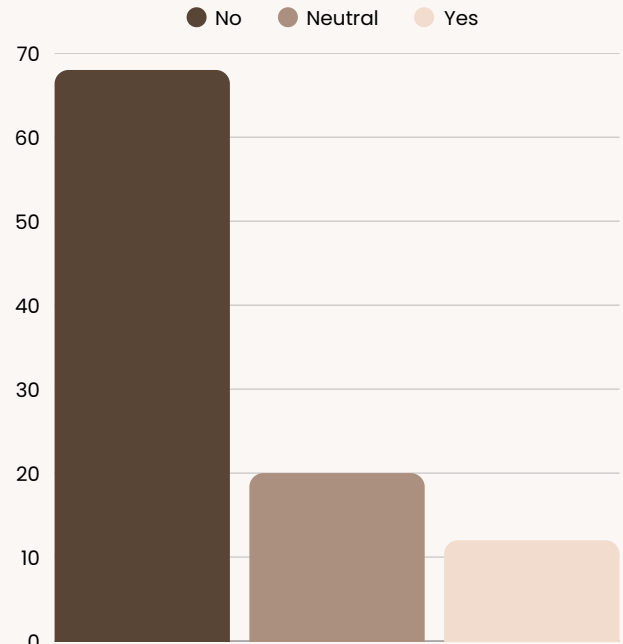
NAVIGATE THE FUTURE

As artificial intelligence continues to shape the future of coding, developers are not simply being replaced—they are evolving. Figure 2 to the right presents data from the 2024 Stack Overflow Developer Survey, where over 36,000 developers were asked whether they believe AI is a threat to their current job. Nearly 70% responded no, while a smaller portion remained unsure or said yes. This chart reflects a growing confidence among developers in their ability to adapt and thrive alongside AI.

This attitude highlights an important perspective among developers: AI is a powerful tool to enhance work, not something that will replace developers. Approaching innovation without fear highlights resilience and a mindset geared toward growth. Preparing for a career in this evolving field means staying open to learning about AI and how to work with it, which will be key to navigating the changes ahead and making the most of emerging opportunities.



Figure 2: Do you believe AI is a threat to your current job?



Source: Stack Overflow 2024

Contrary to the common fear that AI will make junior developers obsolete, recent research shows that they often benefit the most from working alongside AI tools. In a study conducted in a professional software development setting, junior coders experienced greater productivity gains from AI assistance than the senior developers. This is likely because newer developers are more open to using AI, less focused on its imperfections, and more willing to integrate it into their workflow.⁶ Rather than replacing junior talent, AI is amplifying it—helping them learn faster, build confidence, and contribute meaningfully early in their careers. The message is clear: junior developers shouldn't fear AI, they should embrace it as a powerful ally for growth and advancement.

“The world we live in seems to spin a little faster every day... Our workplaces are exposed to the same force. Those who can keep up the pace remain successful... those who cannot keep up are left behind.” ⁷

The pace of change in today’s world is relentless, and workplaces feel this pressure just as much. Those who keep up will succeed, while those who don’t risk falling behind. Research shows that AI is not here to replace developers but to work alongside them.⁸ Understanding how to collaborate effectively with AI and staying adaptable will be essential. Those who remain curious and open to change will be best positioned to thrive in the future of work with AI.

NOTES

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3. McKinsey Global Institute, Generative AI and the Future of Work in America, July 2023, <https://www.mckinsey.com/mgi/our-research/generative-ai-and-the-future-of-work-in-america>.
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ABOUT THE AUTHOR

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Bio: Parker Hendrix is set to begin the Information Systems program at the Marriott School of Business in Fall 2025. He currently works as a web developer for BYU Production Services, where he builds and maintains websites for university departments. Parker is passionate about software development and excited to pursue a future career in business and technology. He is an active member of the BYU Association for Information Systems (AIS) and the BYU Pickleball Club. Outside of school and work, he enjoys spending time with his wife and playing pickleball with her.



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